

Boeing's Safety-First Turnaround: Using a Balanced Scorecard to Rebuild Quality, Trust, and Financial Recovery

This report analyzes Boeing's safety, quality, and trust crisis through the M9 Balanced Scorecard framework, arguing that the company must rebuild operational control, supplier accountability, employee reporting trust, and regulator confidence before pursuing production growth and long-term financial recovery.

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EXECUTIVE SUMMARY

The strategic execution challenge facing Boeing Company is not only in having gone through several safety incidents on its aircrafts. The greater problem lies in the fact that there is a mismatch between the safety, quality, supplier management, reporting, and manufacturing management systems of Boeing Company and its proclaimed emphasis on safety and engineering proficiency. All the incidents, including the crash of the 737 Max series of aircraft, the Alaska Airlines incident involving the 737-9 airplane door plugs, FAA audit results, quality issues with the suppliers, fears of reporting by employees, and the financial strain, show that Boeing Company has failed to consistently translate the "safety first" policy into actual behavior.¹ The incidences of safety and quality issues are summarized in [Appendix A](#).

As Escape Velocity Partners states, the Balanced Scorecard for Boeing should be based on safety first. The Board should give its blessing to five correlated initiatives: to restore the leadership and governance through engineering, operations, and safety; to incorporate Spirit AeroSystems under a single Boeing-controlled quality system; to limit or curtail production expansion until quality gates are achieved; to start using a Balanced Scorecard Dashboard at the Board level; and to make more weight of executive compensation depend on safety, quality, and recovery of operations, rather than on the speed of production and other short-term metrics.

M9 Balanced Scorecard is the proper execution vehicle for Boeing, as the company's problems are multi-dimensional. Recovery of its finances depends upon regained trust from the regulators, airlines/customers, stricter internal manufacturing control, improved supplier quality, and a corporate culture that makes its employees aware of safety procedures and free to report any issues. Using only financial indicators would not allow us to tell if Boeing became safer. Balanced Scorecard will be capable of linking leading indicators, including training, decrease in defects, reports about safety issues, and FAA audits, to lagging measures of deliveries, costs of rework, trust of customers, and profits. The Strategic Map and Dashboard are presented in [Appendix B](#) and [Appendix C](#).

The recommended implementation plan will adopt a "fix control, then increase production" approach. Within the first 90 days, Boeing must develop its strategy on a safety-first basis, establish Board-level scorecard governance, stop unsafe production rate increases, empower stop-the-line capability, and create a baseline of safety and quality metrics. In Months 3 through 6, Boeing must integrate Spirit into one quality system controlled by Boeing, standardize inspections and documentation, improve supplier audits, and introduce part-level accountability. In Months 6 through 12, Boeing must run the full Balanced Scorecard dashboard, demand that production rate increases go through quality gates, and restructure executive compensation. A roadmap for this implementation plan is included in [Appendix D](#).

A safety message is not what is recommended in this report. The report calls for safety to be an operating system. To gain production, Boeing must first prove its safety and quality and not try to build trust later.

CONSULTING ISSUE AND CLIENT CONTEXT

Boeing is one of the key strategic aerospace firms operating in the U.S. This firm has operations in commercial airplanes, defense, space, security, and global services. The commercial aircraft segment of Boeing is mostly a duopoly with Airbus, and thus Boeing has high market significance and dependency on its customers. Market power cannot save Boeing from execution problems. There is a lot of dependence on the part of airlines, regulatory bodies, travelers, workforce, suppliers, and stockholders for Boeing to perform safely and consistently.²

¹ Agnihotri, A., & Bhattacharya, S. (2024). Boeing: A strategy map perspective on resolving a safety crisis. Ivey Publishing. The case describes the 737 MAX crashes, the Alaska Airlines 737-9 incident, FAA concerns, supplier issues, production pressure, employee fear of retaliation, and leadership questions as connected strategic issues.

² Agnihotri, A., & Bhattacharya, S. (2024). Boeing: A strategy map perspective on resolving a safety crisis. Ivey Publishing. The case identifies Boeing's business segments as commercial airplanes, defense, space and security, and global services, and describes Boeing's position in the commercial aircraft market.

The problem that faces the company right now is that the Boeing Board needs to figure out how to restore their safety, control, and trust after multiple incidents regarding poor quality and safety on their 737 MAX, and how leadership, discipline in production, and their integration of Spirit AeroSystems need to be handled.

The preceding recommendation for Boeing was to first repair safety, trust, and control, and then increase production. The recommendation consisted of four main steps: rebuilding leadership with an emphasis on engineering and operations; integrating Spirit back into Boeing's quality system; reducing production rates until quality gates were satisfied; and measuring performance via a Balanced Scorecard. This consulting report amplifies this recommendation with a focus on M9: Building a Balanced Scorecard as the core strategy execution module.

The central claim in this report is that the crisis at Boeing is an example of a strategy implementation problem rather than a communication problem or a transient quality issue. Boeing has continually declared that safety and quality are important, but it does not seem as though the operating processes, incentives, supplier network, and culture within Boeing have been consistent with such a commitment. It follows that the solution for a consultant would need to revolve around how Boeing manages safety.

METHODOLOGY AND SOURCES

In the current report, a systematic approach for consulting has been adopted to find out Boeing's strategy implementation problem and provide a strategy implementation plan using the M9 Balanced Scorecard module. First and foremost, the core problem faced by Boeing was identified, which states that the safety of Boeing, its quality, management of suppliers, its reporting system, and its production processes do not align with its commitment towards safety.

This study is based mostly on secondary research and strategic analysis within the course. They comprise Boeing official sources, sources from the FAA and NTSB regulatory bodies, information about 737 MAX crashes, 737-9 door plug incident of Alaska Airlines, Spirit AeroSystems quality problems, Boeing financial performance, and course material M9: Building a Balanced Scorecard. These sources have been selected to relate Boeing's public safety and quality problems to the strategic execution principles discussed in the course.

Various strategic frameworks are applied in the report before proceeding to the Balanced Scorecard framework. The PESTEL framework is utilized to analyze the influence of politics, economy, society, technology, environment, and legislation on Boeing. Porter's Five Forces framework is utilized to analyze the competitive environment of the commercial aviation production industry. The VRIO framework is utilized to analyze the resources available at Boeing and whether the resources available, such as engineering skills, brand value, supplier network, and regulatory relations, provide sustainable advantages to the company.

Following the strategic diagnosis, the Balanced Scorecard is selected as the primary execution module in the report as per module M9: Building a Balanced Scorecard. It is because the success of the turnaround process at Boeing cannot be determined by short-term performance indicators such as financial performance or production output. The Balanced Scorecard will help to determine how leading indicators, such as safety training for employees, safety incidents, defect rate by suppliers, rework, accuracy of documentation, and FAA audit performance, relate to lagging indicators.

Alternatives that have been considered and rejected prior to making the final recommendation are also included in the report. They include focusing mainly on public relations, improving performance through higher production, solving problems only with respect to the suppliers' quality, and taking leadership replacement as the key solution. The above alternatives are rejected as they are unable to solve the overall problem in the system.

These tools are used for validating the business rationale for the recommendation. In this regard, the financial model will compare the cost of the safety-first approach with the possible financial losses resulting from quality escape, delay, rework, restrictions imposed by the FAA, and loss of customer trust. The sensitivity analysis will show that the success of Boeing's recovery is dependent on various assumptions, like defect rates, rework costs, delivery certainty, etc.

The methodology also has its weaknesses. The information used for the report was publicly available and not from Boeing's internal databases, meaning that accurate defect rate figures, training results, employee reports, and supplier performance cannot be provided. As a result, some financial and operational assumptions had to be estimated. But the information that we have is enough to conclude that Boeing's problem is not just about communication or poor-quality products. Boeing's problem is about executing its strategy with a safety-first approach to operations.

STRATEGIC FRAMEWORK ANALYSIS

Before the implementation of the M9 Balanced Scorecard module, an assessment of the current state of Boeing needs to take place through external and internal strategy analysis. The PESTEL and Porter's Five Forces models outline the external influences on Boeing, and the VRIO model and value chain analysis outline the internal factors influencing Boeing's ability to implement the strategy. As can be seen from this analysis, the crisis at Boeing is not a simple PR or manufacturing issue, but rather a strategy implementation issue.

PESTEL ANALYSIS

POLITICAL

Boeing operates in an intensely political space owing to the connection between aviation safety and public confidence, infrastructure, defense, and regulation. The FAA, NTSB, Congress, and other international regulators affect Boeing's capability to certify airplanes, manufacture more airplanes, deliver airplanes, and repair its image. Political pressure has escalated following the 737 MAX crashes and Alaska Airlines' 737-9 door plug problem since regulators and politicians started scrutinizing Boeing's safety culture and manufacturing process more closely. Thus, regulatory trust becomes one of the key strategic resources for Boeing. Boeing must rebuild its reputation not just through delivery increases but through establishing its safety and quality control system.

ECONOMIC

Boeing faces strong economic pressure because its financial recovery depends on production volume, aircraft deliveries, cash flow, airline demand, and cost control. However, the company's economic challenge is complicated by rework costs, production slowdowns, quality problems, supplier issues, legal exposure, and customer concerns. The financial temptation is to increase output quickly to restore revenue but doing so before quality systems are reliable could create even higher long-term costs. For Boeing, economic recovery depends on operational recovery. The company must reduce defects, delays, and rework before production growth can become financially sustainable.

SOCIAL

Confidence in the company Boeing has been eroded since the safety of aircraft concerns passengers, customers of the airline companies, the employees themselves, regulators, and the public. The brand of Boeing used to stand for excellence in engineering, but safety and quality failures led to erosion of trust. There is also a social and cultural problem inside Boeing, since the employees should feel safe enough to report the problems and stop working in case of any quality problems. If they fear being punished or think that speed of production takes precedence over safety, then Boeing's culture will keep hiding its strategy.

TECHNOLOGICAL

The Boeing business operates within an industry characterized by the importance of technology, engineering expertise, manufacturing processes, digitalization of information, traceability of components, and quality control systems for competitive advantage. Aircraft are complicated products that have thousands of parts and several suppliers. Minor issues in the process could result in major safety problems. For Boeing, its technological issue is not just about creating state-of-the-art

aircraft; it must ensure that all aspects, such as engineering knowledge, manufacturing processes, inspection, and suppliers' information, integrate with each other.

ENVIRONMENTAL

The environmental pressure on Boeing is seen in terms of demands from airlines for more fuel-efficient aircraft, emissions, sustainable considerations, and innovative aspects in aviation. Though the present dilemma of Boeing is concerned with safety issues and quality issues, it is imperative to recognize the role played by environmental expectations, as airlines will look for aircraft that are safe, efficient, reliable, and sustainable in the future. It will be difficult for Boeing to compete with Airbus and other upcoming aerospace companies if it focuses only on solving its safety concerns.

LEGAL

Boeing is under significant legal risk in case of aircraft accidents, safety concerns, investigations, manufacturing issues, customer complaints, and supply chain quality problems. Legal risk is heightened if safety problems seem to be more systemic than individual cases. The legal context serves to emphasize the need for documentation, inspection data, proof of training, corrective measures, and executive responsibility. If Boeing does not show that it has good control systems in place, legal and regulatory risk will keep hurting the company.

PORTER'S FIVE FORCES ANALYSIS

COMPETITIVE RIVALRY: HIGH

Competitive rivalry in the production of commercial aircraft is high since Boeing and Airbus have cornered the market in large commercial aircraft. Airline companies evaluate aircraft depending on factors like safety, reliability, delivery dates, fuel efficiency, cost of operations, and maintenance. Competitiveness in this situation is very harmful to Boeing since whenever there are any quality problems, Airbus will benefit from the lack of trust in Boeing's capability to produce quality planes. Quality problems are dangerous to Boeing since not only will they affect current performance, but also future performance.

THREAT OF NEW ENTRANTS: LOW

The potential for new entrants is low due to the large amount of money required to develop a commercial aircraft manufacturing business, along with engineering skills, government approval, suppliers, a long time to develop, and customers worldwide. Low potential of entry does not safeguard Boeing from failing in its implementation. It might be difficult for a new competitor to enter the marketplace of Boeing, but there will be repercussions from the customers and government in terms of orders and production limits.

BARGAINING POWER OF BUYERS: MODERATE TO HIGH

The purchasing power of airline customers is high because they make purchases of costly aircraft in bulk. The airlines cannot change their supplier of goods immediately; however, they can postpone their orders, bargain with the supplier, and even choose to buy from Airbus in the future. The problem of high buyer power for Boeing becomes more pronounced when its reliability comes into question.

BARGAINING POWER OF SUPPLIERS: HIGH

The supplier's power is strong since the production of an aircraft requires the involvement of specialized suppliers, long-term agreements, technical expertise, and sophisticated parts. The significance of Spirit AeroSystems is even greater because the quality of the supplier's products is linked to the overall quality of Boeing aircraft. Boeing cannot isolate its

suppliers' problems from Boeing's problems since the airlines, authorities, and customers are accountable only to Boeing regarding the aircraft.

THREAT OF SUBSTITUTES: LOW TO MODERATE

Direct substitution in the case of commercial planes for long-distance flights is very limited. Substitution may come in the form of using different plane types, delay in replacement of fleet, high-speed rail in certain cases, and prolonging the use of existing planes by airlines. But the problem faced by Boeing is not that airlines do not need planes anymore, but that they switch from Boeing to Airbus.

VRIO ANALYSIS

BOEING'S ENGINEERING EXPERTISE

The value of Boeing's engineering expertise lies in the fact that it takes very sophisticated technological skills to create planes. The rarity lies in the fact that not many firms have the capacity to design and manufacture commercial airplanes in a global marketplace. Imitability is also low since engineering expertise is dependent on years of experience, skilled personnel, certification, relationships with suppliers, and technological knowledge. The only problem here is whether Boeing is structured to maximize the value of such an asset.

BOEING'S BRAND AND REPUTATION

The Boeing brand has significant value for many stakeholders, including airlines, regulatory agencies, airline passengers, and government institutions, who see Boeing as one of the world's leading aerospace companies. It is unique as there are very few aerospace brands that have such a long history and market presence as that of Boeing. It is hard to duplicate because of the way reputation is created over time. But the fact is that the Boeing brand has been tainted by the recurring safety and quality problems.

BOEING'S SUPPLIER NETWORK

The supplier network at Boeing is important since it is necessary to have specialized suppliers to produce large aircraft. This source of competitive advantage is also quite rare since few companies can organize such a supply chain for their aerospace business. It is not easy to imitate since supplier connections, tooling, engineering requirements, and certification all take time to arrange. Boeing's supplier network has turned into a weakness for Boeing because of poor quality control.

REGULATORY RELATIONSHIPS AND CERTIFICATION KNOWLEDGE

The Boeing Company's experience in certification with the FAA and international agencies is important since companies in this industry must meet high levels of safety and regulatory compliance. This type of resource is uncommon and difficult to replicate because it requires extensive experience in the certification of complicated aircraft projects. The resource has become weaker as regulators' trust in Boeing has declined.

MANUFACTURING SCALE

The manufacturing scale of Boeing is significant because of its ability to manufacture large planes for airlines worldwide. The reason why it is unique is that no other company can match its manufacturing capabilities. Scale is hard to replicate since it involves having facilities, workforce, suppliers, engineering, and authorization. On the other hand, scale can become a threat when a company increases its production without adequate quality systems. Thus, the manufacturing scale of Boeing should be viewed as an asset when properly controlled.

VALUE CHAIN ANALYSIS

INBOUND LOGISTICS AND SUPPLIER MANAGEMENT

The value chain for Boeing starts with suppliers, parts, fuselage sections, documentation, inspection, and transfer. Supplier quality is one of the most critical components of the value chain for Boeing, as poor quality may lead to issues during final assembly and delivery of the final product. Spirit AeroSystems play an important role in this matter as Boeing's final product depends on supplier quality. Value chain analysis reveals that Boeing requires better supplier audits, defect tracking collaboration, part traceability, and quality gates controlled by Boeing.

OPERATIONS AND MANUFACTURING

This is where Boeing takes raw materials, engineering design, manpower, and suppliers' efforts and turns it all into planes. Operations are where pressure on the production line may turn into a safety hazard. If people such as mechanics, inspectors, and management are forced to move planes rather than take care of quality issues, the value chain will generate defects instead of value. Boeing's operations require standardization, better training, more documentation, stop-the-line authority, and quality gates to ensure that planes cannot go ahead until problems are solved.

OUTBOUND LOGISTICS AND DELIVERY

The outbound logistics of Boeing entail conducting the final inspection of the aircraft, preparing for the delivery, handing over to the customer, and being ready for regulations. Delivery is not only an achievement for sales but also an indication that the aircraft successfully passed the safety and quality process. In case Boeing delivers aircraft with quality problems, it will damage its customer confidence, regulatory trust, and financial value.

MARKETING, SALES, AND CUSTOMER TRUST

The selling process of Boeing relies greatly on the trust of customers. The airlines do not purchase planes purely based on price. They make decisions based on safety, reliability, operating costs, dependability of delivery, and continued support service. Marketing cannot resolve the issue of trust for Boeing without the right operational system.

SERVICE AND AFTER-SALES SUPPORT

Service operations at Boeing comprise maintenance services, technical support, problem-solving, communications, and relationships with clients. Service takes on greater importance if there is any concern on the part of the client related to issues of safety and reliability. Good service may aid Boeing in rebuilding its image if it solves the underlying causes of the quality problems. Service must provide feedback on manufacturing, supply management, training, and dashboard reporting.

SUPPORT ACTIVITIES: LEADERSHIP, HR, TECHNOLOGY, AND PROCUREMENT

Boeing's supporting functions play a crucial role in the situation at hand. The company's leaders need to set priorities, the HR function needs to be involved in training and reporting culture, technology will help with documentation and traceability, and procurement will manage the suppliers' expectations. In case Boeing's supporting functions incentivize speed of production rather than the safety record, the whole value chain is out of alignment.

VALUE CHAIN CONCLUSION

The value chain analysis indicates that the problems Boeing is facing as far as its safety and quality concerns could arise at a few different but interrelated stages, including the handing over from suppliers, paperwork, inspections, manufacturing, reporting, distribution, and support. This is why the Balanced Scorecard approach is needed, since a single measure cannot reflect all the complexity of the situation.

OVERALL FRAMEWORK CONCLUSION

All four strategic frameworks lead to the same conclusion. According to PESTEL analysis, Boeing finds itself under the pressure of regulation, law, society, and economics. As per the Five Forces analysis, Boeing works in a highly risky environment in which customer and supplier power is crucial. VRIO analysis reveals that Boeing possesses valuable resources that become weak due to misalignment on safety and quality issues. Value chain analysis proves that the problems of Boeing are interrelated at the levels of supplier management, manufacturing, inspecting, delivery, and support.

Thus, the proposed solution cannot be limited only by PR and change of leadership or even production restoration. Boeing needs an execution system that will ensure translating its "safety-first" strategy into measurable action. That is the reason why M9: Building a Balanced Scorecard is the right module for the report.

STRATEGIC OPTIONS CONSIDERED

In deciding upon the recommendation to be presented, the report reviewed different strategies that could be considered by Boeing to make their company whole again. Some of these were to continue with the current strategy, improve the auditing and contractual relationship with Spirit but leave Spirit separate, have a third-party/FAA oversight system, or re-integrate Spirit, considering a more safety-minded Balanced Scorecard approach.

In comparing these alternatives, the current arrangement will be the quickest short-term solution; however, it does not solve the underlying problem of system failure. Enhanced audit capabilities and contract provisions will lead to better oversight, but the problem of fragmented accountability will still exist. Depending on external oversight from third parties or even the FAA, it can help restore faith, but Boeing cannot outsource its quality management responsibility to anyone. The best long-term solution is the integration of Spirit into a safety-first Balanced Scorecard Reset, since it solves all these issues simultaneously. A complete comparison is provided in [Appendix E](#).

WHY M9: BUILDING A BALANCED SCORECARD FITS BOEING

Balanced Scorecard is meant to help translate strategic thinking into actions that are measurable. As per the M9 module, the scorecard will help relate goals and metrics to the strategy in an employee-friendly manner. The importance of the balanced scorecard lies in its ability to enable the leadership to state the theory of value creation and align the organization's employees, processes, and functions towards the objectives. In the case of Boeing, this aspect is relevant because the challenge of the firm is not the absence of broader objectives. Boeing already uses terms like safety, quality, innovation, and trust.

Step-by-step process from the M9 module on construction of the Balanced Scorecard includes defining the strategy of the firm, constructing a strategy map, measuring the map, and defining the accountability of the managers. These steps have relevance to the situation at Boeing Company, and specifically to its turnaround. The first step will be to define Boeing Company's strategy, namely, its strategy should be safety-first rather than production-first recovery. The second step will be to construct a strategy map in such a way as to show that safety culture and employee capability create internal reliability that will restore regulators' and customers' confidence and, therefore, lead to financial recovery. The third step will be to measure Boeing's strategy through those measures that can talk about improvements in safety and quality before any improvement in financial results becomes visible. Boeing's strategy map is provided in [Appendix B](#).

The classic Balanced Scorecard consists of four perspectives: financial perspective, customer perspective, internal business processes perspective, and learning and growth perspective. In the case of Boeing, these perspectives must be modified to match the conditions of the aviation safety industry. The four suggested perspectives are:

1. Learning and Growth Perspective: Safety culture, employee training, engineering expertise, reporting integrity, and stop-the-line.

2. Internal Processes Perspective: Manufacturing quality, supplier management, defect minimization, inspection integrity, documentation, and parts traceability.
3. Customer and Regulator Trust: FAA confidence, airline confidence, auditing excellence, on-time deliveries, and openness.
4. Financial Perspective: Lower rework and crisis costs, sustainable production growth, improved cash flow, fewer delays, and shareholder value.

This adaptation is necessary since Boeing's customers as well as regulators are equally important stakeholders. In any other industry, the customer perspective is all about customer satisfaction, loyalty, or preference for a brand. In the case of commercial aviation, there can be no talk of customer trust without regulatory trust. The airlines purchase airplanes from Boeing, but regulators decide about certification, inspection, delivery, and trustworthiness of the airplanes in the eyes of the public. Hence, the scorecard for Boeing needs to include FAA and other regulators, along with the airline customers.

M9 MODULE WALKTHROUGH: FROM STRATEGY TO ACCOUNTABILITY

The M9 Balanced Scorecard tool is used at Boeing by following a structured process of implementation. The first process would involve defining the strategy. In the case of Boeing, the strategy statement should be: "solve safety and quality control issues and then ramp up production." This is important because Boeing's recovery cannot rely only on production speeds. It is important to consider production growth as something that Boeing has earned based on its other systems.

The second stage will involve developing a strategy map for the company. The strategy map for Boeing should incorporate an effect-to-cause relationship beginning with capabilities and culture of safety among the employees, to quality internal processes, then on to trust of the customers and regulators, and lastly to financial recovery. This is crucial because the performance of Boeing is lagging behind the results, which are dependent on prior improvements in other areas.

The third step involves choosing measures of performance for each scorecard point of view. Under the Learning & Growth viewpoint, Boeing needs to measure completion of training, reporting on safety by employees, stop-the-line activity, and reporting trust of employees. Under the Internal Process viewpoint, Boeing needs to measure defects by suppliers, rework, accuracy, completion of quality gate, and inspection results. Under the Customer and Regulatory Trust viewpoint, Boeing needs to measure FAA audit results, airline complaints, delivery acceptance, and open corrective actions. Under the Financial Recovery viewpoint, Boeing needs to measure rework cost, delivery reliability, cash flow effect, and production increases.

Accountability is the fourth step that needs to be taken. There must be an assigned executive owner for each of the perspectives of the scorecard. In addition, the Board must review the entire scorecard on a quarterly basis, while the senior management must review it monthly.

Step five involves connecting the scorecard with action. The Balanced Scorecard ought to drive Boeing's big turnaround efforts, like Spirit integration, quality gate reviews, production rate assessments, whistleblower protection programs, training, supplier audit programs, and executive compensation plans. This would transform M9 into an operating system rather than a measurement system.

Finally, the scorecard needs to be reviewed and revised regularly. Boeing should increase targets with improved performance levels, but the company should never delete any safety or quality metrics when it faces financial difficulties. The goal of the M9 module is to ensure that Boeing maintains its link between strategy and behavior, Board monitoring, and value creation.

DIAGNOSIS: BOEING'S CRISIS AS AN EXECUTION-SYSTEM FAILURE

THE CRISIS IS SYSTEMIC, NOT ISOLATED

The recent incidents involving Boeing include the crashes of the 737 MAX aircraft in 2018 and 2019, the grounding of the 737 MAX aircraft, various quality issues, and the 737-9 door plug incident with Alaska Airlines in January 2024. It is apparent from the case data that these incidents did not occur randomly but rather because of certain issues in the above areas.³

However, the incident at Alaska Airlines involving the door plug takes on additional significance as it happened following Boeing's efforts to regain its reputation after the 737 MAX tragedies. The airplane involved was new, the incident resulted in rapid decompression, and the investigation authorities discovered that critical hardware components relating to the door plug were missing. The NTSB would later conclude that the most likely reason for the incident to have taken place was Boeing's failure to properly train and guide the manufacturing personnel regarding part removal procedures.⁴

This is the reason a Balanced Scorecard is needed. Boeing cannot fix its systemic problem through any one corrective measure alone. Boeing requires an approach that integrates people, process, trust, and financial results. If Boeing looks only at the result of finance, then it misses the underlying factors that allow a quality escape to happen. On the other hand, if Boeing looks only at a particular instance at a certain plant, then it fails to look at the bigger picture of culture and supplier control.

PRODUCTION PRESSURE UNDERMINED SAFETY DISCIPLINE

Another reason for the problem is the production pressure. Once the 737 MAX was grounded, there was pressure on Boeing to boost production, restore the deliveries, gain back market share from Airbus, and improve the business performance. Yet, according to the case, there were fears that employees were pressured to continue the movement of planes along the assembly line, rather than stop resolving the issue. The strategic problem is not in the production growth itself. Boeing will need to produce and deliver planes on a large scale. What becomes a problem is the production growth outpacing the organization's quality control capabilities.⁵

This creates an excellent example of execution tension where the organization requires revenue and volume of deliveries, yet the system cannot be scaled up without addressing the issue of capability first. While a production-based approach to recovery may appear more cost-effective in a shorter period, it can destroy much more value if defects, grounding, customer frustration, regulations, and legal actions take place. In this way, Boeing must use the Balanced Scorecard and make production growth dependent on quality readiness, not vice versa. [Appendix F](#) shows the suggested decision-making criteria of the Board approving production increases.

SUPPLIER CONTROL WAS TOO WEAK

Spirit AeroSystems is critical to this case since Boeing does not make the 737 MAX by itself. Spirit made important components of the fuselage and was a part of the chain for the doors plug supply.⁶ The reason why all this is strategically important is that the quality of the supplier cannot be considered separately from the Boeing quality, as the aircraft will always remain a Boeing one for the airline, regulator, and passenger, no matter who has produced some parts of the plane.

The old supplier model led to accountability being split up. This was because when Spirit manufactured a part, Boeing assembled and delivered the aircraft, while the FAA was responsible for the entire process. Thus, the reintegration of Spirit activities by Boeing makes perfect strategic sense, although the acquisition of the company alone may not be sufficient. Instead, what Boeing needs is one quality control system that it controls. This will involve standards, documentation,

³ Agnihotri, A., & Bhattacharya, S. (2024). Boeing: A strategy map perspective on resolving a safety crisis. Ivey Publishing.

⁴ National Transportation Safety Board. (2025, June 24). Boeing's inadequate "training, guidance and oversight" led to mid-exit door plug blowout on passenger jet. <https://www.nts.gov/news/press-releases/Pages/NR20250624.aspx>

⁵ Agnihotri, A., & Bhattacharya, S. (2024). Boeing: A strategy map perspective on resolving a safety crisis. Ivey Publishing. The case discusses deadline pressure and employee concerns that production-line pressure discouraged stopping work to fix problems.

⁶ Agnihotri, A., & Bhattacharya, S. (2024). Boeing: A strategy map perspective on resolving a safety crisis. Ivey Publishing. The case's supplier issues section discusses Spirit AeroSystems, Boeing-Spirit communication problems, quality-control failures, and the door plug handoff process.

inspection, training, and accountability for all the parts moving from the supplier to the final assembly. Strategic options that resulted in the above recommendation are discussed in [Appendix E](#).

EMPLOYEE CULTURE AND TRAINING WERE INSUFFICIENT

Yet another root cause is the safety culture and preparedness of Boeing's employees. According to the case information, Boeing's employees lacked awareness of their safety responsibilities and were afraid of potential retaliation. This is particularly harmful in a high-reliability setting. In the production of aircraft, it is critical to have all frontline employees, mechanics, inspectors, engineers, and managers spot any issues prior to those issues reaching the clients.⁷

This is the reason why the Learning and Growth approach must be at the bottom of Boeing's strategy map. For most organizations, learning and growth may come off as a soft issue. But for Boeing, learning and growth are mandatory issues for safety purposes. Learning and growth issues, such as training, reporting of trust, engineering competency, and stopping the line authority, are hard control issues.

REGULATOR AND CUSTOMER TRUST BECAME A STRATEGIC ASSET AT RISK

FAA's heightened oversight, production limitation, and audit results revealed that regulatory trust acted as a strategic constraint. In January 2024, following the incident involving Alaska Airlines, the FAA indicated its refusal to allow further production increase for the 737 MAX aircraft until all quality control issues were sorted out.⁸ Boeing was tasked to come up with an action plan for quality control and production problems.⁹

For Boeing to be able to survive, it is not enough to assure the investors that there will be an increase in the number of production units. They need to persuade the regulators that their production process is under control and convince the airlines that deliveries will be safe and reliable. The customer/regulator point of view needs to evaluate more than just satisfaction.

FINANCIAL RECOVERY DEPENDS ON SAFETY RECOVERY

Boeing's financial performance has been negatively impacted by issues related to quality, slowdown in production, rework, delay in delivery, and cost of crises. According to the exhibit in the cases, Boeing's market capitalization has dropped substantially from 2018 to 2024. This has been stated in the annual report of Boeing in 2024 regarding the impact of safety and quality initiatives on production and operations.¹⁰ The Board must be careful not to compartmentalize financial recovery from safety recovery at Boeing.

It is for this reason that this report recommends that the financial metrics at Boeing should be lagging measures. The measures will continue to form part of the balanced scorecard, but they mustn't be overwhelming measures on the balanced scorecard. When the executives are compensated in terms of deliveries, reduction in costs, or cash flow, there is a danger of inadvertently creating the same pressures that caused the crisis.

⁷ Agnihotri, A., & Bhattacharya, S. (2024). Boeing: A strategy map perspective on resolving a safety crisis. Ivey Publishing. The case discusses FAA findings that employees did not always understand safety roles and feared retaliation for raising concerns.

⁸ Federal Aviation Administration. (2024, January 24). FAA halts Boeing MAX production expansion to improve quality control, also lays out extensive inspection and maintenance process. <https://www.faa.gov/newsroom/faa-halts-boeing-max-production-expansion-improve-quality-control-also-lays-out-extensive>

⁹ Federal Aviation Administration. (2024, May 30). FAA continues to hold Boeing accountable for implementing safety and production quality fixes. <https://www.faa.gov/newsroom/faa-continues-hold-boeing-accountable-implementing-safety-and-production-quality-fixes>

¹⁰ The Boeing Company. (2025). The Boeing Company 2024 annual report. <https://www.boeing.com/content/dam/boeing/boeingdotcom/company/annual-report/2024/2024-annual-report.pdf>

STRATEGY MAP: HOW BOEING CREATES VALUE THROUGH SAFETY

The proposed strategy map (**Figure 1**) below follows a cause-and-effect chain:



Figure 1

It is very straightforward. Boeing will not be able to achieve internal quality improvements without first training and empowering its staff while also ensuring that they have sufficient protection when they voice their safety concerns. The processes will not become reliable for Boeing without first improving its inspections, documentation, suppliers' control, and parts traceability. Trust from the customer and regulatory authorities will not be restored without improving the performance of the internal processes and making them visible. This cause-and-effect relationship is depicted in [Appendix B](#).

LEARNING AND GROWTH: REBUILD THE SAFETY CULTURE FOUNDATION

Learning and Growth is the first level in the strategy map. Boeing should develop its objective at this level by rebuilding the safety culture and the responsibility of the employees. The major initiatives at this level should include mandatory safety and quality training, better training of mechanics and inspectors, management training, stop-the-line authority, better whistleblower protection, and employee involvement teams.

Recommended metrics are included in **Figure 2** to the right:

Most critical among all these changes would be for Boeing to view the reporting of safety information by employees not as a problem but as an early warning signal that indicates whether there are systemic problems at work. Although an increase in safety reporting may appear to have a negative impact on operations in the short term, it could be a good thing.



Figure 2

INTERNAL PROCESSES: BUILD QUALITY GATES THAT ACTUALLY STOP DEFECTS

The second level is internal processes. Boeing's strategic target should be improvement in the manufacturing quality of the plane before delivering it. This includes inspection of the suppliers, removal documentation, tooling control, parts handling, process conformity, and rework confirmation.

Recommended metrics are found in **Figure 3** to the right:

The most important suggestion in terms of process improvement is that of implementing quality gates. Quality gate means a gate through which the project cannot move without meeting certain quality criteria. In the case of Boeing, quality gates should be implemented at Spirit/Boeing Hand-off, Fuselage Acceptance, Final Assembly, Rework Completion, Pre-Delivery Inspection, and FAA Certification Readiness.

CUSTOMER AND REGULATOR TRUST: PROVE THE TURNAROUND EXTERNALLY

The third level is Customer and Regulator Trust. Boeing's strategic goal is to reestablish trust with regulators, airlines, and customers. The only way to regain trust is by delivering good performance, not just making commitments.

Recommended metrics are found in **Figure 4** to the right:

This information needs to be provided to the Board every quarter. If there are any open investigations by regulators, any defects from suppliers, or any complaints from airlines, Boeing should not expand its production capacity, no matter the financial considerations.

FINANCIAL RECOVERY: GROW ONLY AFTER THE SYSTEM IS SAFE

The fourth level is Financial Recovery. Boeing's strategic objective should be to stabilize financial performance through safer production, fewer defects, lower rework costs, and more reliable deliveries.

Recommended metrics are found in **Figure 5** to the right:



Figure 3



Figure 4

The financial recovery will be an outcome, not a starting point. Boeing needs to monitor its financial results, but these financial results need to be evaluated in the context of safety scorecards. If there is an increase in production that does not go along with improvements in quality metrics, then this is not really a recovery.

RECOMMENDATIONS

RECOMMENDATION #1: ADOPT A BOARD-LEVEL SAFETY-FIRST BALANCED SCORECARD

Boeing needs to develop a formal Balanced Scorecard dashboard at the Board level, which will be reviewed monthly by the company's senior management team and on a quarterly basis by the Board. The dashboard will measure progress in relation to Learning and Growth, Internal Processes, Trust of Customers and Regulators, and Financial Recovery. Management will have to provide explanations for both meeting the goals and not meeting the goals.

This recommendation will help to overcome this long-standing problem within Boeing. This will make their commitment to safety evident via a dashboard. This will ensure that the management cannot choose only to highlight their success around finance and not quality.

RECOMMENDATION #2: INTEGRATE SPIRIT AEROSYSTEMS INTO ONE BOEING-CONTROLLED QUALITY SYSTEM

Spirit must be incorporated by Boeing into a system of quality standards and training that is common to all. The objective here is not mere ownership but rather control. The December 2025 completion of the acquisition of Spirit AeroSystems by Boeing signals that the strategic question now is not acquisition but rather integration.¹¹ Overall recommendations are found below in **Figure 6**:

Since Boeing owns Spirit today, the Board should not assume that everything will be fixed because Boeing purchased the company. The question is whether Boeing can incorporate Spirit into its system without causing the same problems again.



Figure 5



Figure 6

¹¹ The Boeing Company. (2025, December 8). Boeing completes acquisition of Spirit AeroSystems. <https://investors.boeing.com/investors/news/press-release-details/2025/Boeing-Completes-Acquisition-of-Spirit-AeroSystems/default.aspx>

RECOMMENDATION #3: TIE PRODUCTION GROWTH TO QUALITY GATES

Boeing shouldn't pursue production rates independently. An increase in production should occur only once the company has satisfied its safety and quality metrics for some time. This implies that the company needs to set acceptable levels for supplier defects, rework, FAA issues, documentation, training, and deliveries.

The recommendation does not imply that Boeing should have reduced production capacity permanently. There comes a time when Boeing needs to scale up to match the demands of airlines and increase profitability. The company must earn its scaling. "Control first, then scale production" should be the guiding principle of the Board.



Figure 7

RECOMMENDATION #4: REALIGN EXECUTIVE INCENTIVES AROUND SAFETY AND QUALITY

The executives' pay package should be modified in such a way that the weightage of safety criteria and quality criteria is more important than that of production time and financial turnaround. Boeing should add modifiers to its executives' pay packages in case of any significant failure regarding either safety or quality. Recommended incentive categories are included in **Figure 7:**

The purpose is to align executive behavior with the safety-first strategy. If compensation rewards speed more than quality, Boeing risks rebuilding the same system that caused the crisis.

RECOMMENDATION #5: STRENGTHEN EMPLOYEE REPORTING AND STOP-THE-LINE AUTHORITY

Boeing must develop reporting mechanisms for its employees and provide protection to whistleblowers in the interest of employee safety. Boeing needs to make its policy on stopping the line clear, transparent, and enforceable. Employees should know when they must stop, how they can voice their concerns, how these will be investigated, and protection from any reprisal.

Metrics should also measure the volume of reports, the average resolution time of reports, the number of retaliatory complaints, survey results for levels of employee trust, and the number of stop-the-line incidents closed due to confirmed corrective action. High-quality reporting by Boeing employees is a sign of system maturity.

FINANCIAL MODEL AND SENSITIVITY ANALYSIS

The economic rationale for the recommendation is that even though Boeing's safety-first recovery strategy may limit short-term production growth, it minimizes the financial risks arising from repeated quality escapes, rework, delays in delivery, regulatory issues, loss of trust from customers, and crisis costs. In the model, the safety expenditure is not considered a different cost from the financial recovery process, but rather, safety and quality control become the basis for financial recovery.

The model contrasts three different situations: the baseline recovery, the safety-first recovery case, and the downside case. In the baseline recovery situation, the assumption is that Boeing will continue to make gradual improvements but will

continue to have quality disruption. The safety-first recovery case assumes that Boeing will invest earlier in supplier integration, training, quality gates, documentation, and scorecard governance at the Board level. The situation involves a higher initial cost of implementation, but low rework, fewer delays, high reliability of deliveries, and a low likelihood of future regulatory restrictions.

The financial logic of the recommendation is explained by the comparison of three possible recovery scenarios for Boeing: a baseline recovery, a safety-first recovery, and a worst-case production-first scenario. The goal of this comparison is not to predict Boeing's earnings going forward, but to illustrate the impact of different implementation scenarios on costs, risks, production reliability, regulator confidence, and financial recovery. **Figure 8** below provides a summary of key assumptions made for each of the scenarios.

Figure 8

Financial Scenario Assumptions and Expected Impact			
<i>Boeing recovery decision model</i>			
	 Baseline Recovery	 Safety-First Recovery	 Downside / Production-First
 Implementation Cost	Moderate	High in the short term	Low in the short term
 Rework / Delay Cost Risk	Moderate	Lower over time	High
 Production Assumption	Gradual recovery with some continued disruption	Slower near-term ramp-up until quality gates are met	Faster production push before controls are stable
 Expected Long-Term Impact	Some improvement, but recovery remains exposed to quality disruptions	Stronger long-term recovery through fewer defects, better trust, and more reliable deliveries	Higher risk of rework, FAA restrictions, customer distrust, and renewed financial damage
 Key takeaway: The safety-first recovery scenario is financially stronger over time even though it carries higher upfront implementation costs, because it reduces the recurring disruptions that damage cash flow, delivery reliability, and regulatory trust.			

While the safety-first recovery approach may be more costly to implement in the short term, the analysis reveals that the financial implications of the approach are better in the long run due to its ability to mitigate the ongoing causes of value destruction.

Assumptions in the model include the fact that the turnaround for Boeing is not dependent on maximum production rates but rather on minimizing defects and reworks, reliability of delivery, confidence of the FAA, and customer trust. Sensitivity analysis was conducted to identify the assumptions that would have the greatest impact on the model. The key factors include the rate of defects/reworks, the cost per plane of reworks, the duration of production delays, the rate of FAA approval for increased production, and the cost of safety-first reset.

Sensitivity Analysis backs up the recommendations made in the report. Though the safety-first approach may cost more initially, it will be more cost-effective since it will save the company from any rework, disruption in delivery, and another quality scandal. The complete analysis is included in [Appendix G](#).

IMPLEMENTATION PLAN

FIRST 0–90 DAYS: DEFINE STRATEGY, BASELINE METRICS, AND FREEZE UNSAFE SCALING

The first 90 days need to be about control. Boeing should not try to fix everything that has gone wrong right away. Rather, it should outline its “safety first” approach, establish governance, determine benchmark metrics, and make sure that production pressure does not swamp the restart effort.

Key actions:

- Establish a Board Safety and Quality Scorecard Committee.
- Define the safety-first strategy in one sentence: Boeing will earn production growth by proving safety and quality performance first.
- Freeze production-rate increases that are not supported by quality-gate evidence.
- Create a cross-functional Safety and Quality Task Force.
- Assign executive ownership for each scorecard perspective.
- Conduct a baseline audit of training, defect rates, documentation, rework, FAA findings, and supplier quality.
- Expand stop-the-line authority and communicate non-retaliation protections.
- Begin monthly scorecard reporting to the CEO and quarterly reporting to the Board.

The outcome of the process should be visibility for Boeing. Boeing should be aware of its areas of failure, the reliable metrics, the lacking metrics, and the leaders responsible for these areas.

MONTHS 3–6: INTEGRATE SPIRIT AND STANDARDIZE QUALITY PROCESSES

The second phase should focus on supplier and internal process control. Boeing should begin integrating Spirit into one Boeing-controlled quality system and standardizing the procedures that affect aircraft quality.

Key actions:

- Create a Boeing-Spirit Integration Management Office focused specifically on safety and quality.
- Standardized inspection and documentation procedures.
- Implement part-level traceability requirements.
- Create shared defect reporting across Spirit and Boeing.
- Require quality gates at Spirit-to-Boeing handoff.
- Increase supplier audits and make audit findings visible to senior leadership.
- Assign named owners for recurring defect categories.
- Train Boeing and Spirit employees on the same quality-control expectations.

In this phase, the primary deliverable should be accountability. This would enable Boeing to know who is responsible for each process, what defects are being repeated, where there are gaps in documentation, and whether corrective actions mitigate the risks involved.

MONTHS 6–12: LAUNCH FULL SCORECARD GOVERNANCE AND GATE-BASED PRODUCTION

The third phase would be the incorporation of the Balanced Scorecard in decision-making for manufacturing processes. Boeing needs to transition from crisis management to execution

Key actions:

- Launch the full Balanced Scorecard dashboard.
- Require production increases to pass defined quality gates.
- Review scorecard results at Board meetings.
- Tie executive compensation to scorecard outcomes.
- Publish selected safety and quality progress indicators to customers and regulators.
- Use airline feedback and FAA findings to revise internal process targets.
- Track rework cost, delivery reliability, and defect reduction as connected outcomes.

The key to deliverables for this stage is that growth must be well-disciplined. Boeing must grow only when there are positive signs on safety, quality, and trust metrics.

MONTHS 12–24: INSTITUTIONALIZE THE SAFETY-FIRST OPERATING SYSTEM

This last step must ensure that the Balanced Scorecard is not forgotten when the crisis passes. Boeing needs to integrate the Balanced Scorecard into annual planning, executive review, plant management, supplier management, and compensation programs.

Key actions:

- Update scorecard targets annually.
- Conduct independent safety and quality audits.
- Maintain recurring employee safety culture surveys.
- Keep Spirit integration metrics active until performance stabilizes.
- Review scorecard design after major program milestones.
- Use lessons from the 737 program to improve 767, 777, 787, defense, and services operations.

The primary deliverable from this phase would be institutional memory. Boeing must not have to face another crisis to realize the importance of safety discipline.

POTENTIAL CHALLENGES AND MITIGATION

CHALLENGE 1: PRODUCTION AND FINANCIAL PRESSURE

Boeing will feel the need to deliver more, manage its cash flow, and meet the demands of airlines. The problem is that there is always the possibility of prioritizing quantity over quality once again. To prevent this, the board should insist on gate approvals in production.

CHALLENGE 2: INTEGRATION COMPLEXITY WITH SPIRIT

Integration of Spirit will be hard to achieve. The organization needs to integrate systems, workforce, processes, suppliers, and governance in various locations. The problem is that ownership is changed, but the fragmentation of processes persists. To address this challenge, Boeing should establish an integration team having control over the quality management systems, and not only financial and legal integration.

CHALLENGE 3: EMPLOYEE SKEPTICISM

The staff may not be initially inclined to believe in the new reporting procedures for safety concerns. In case the staff had fears of being retaliated against or neglected, it would take time for the culture to change. To alleviate this problem, Boeing should provide feedback to the employees.

CHALLENGE 4: METRIC GAMING

Scorecards will be ineffective when the management distorts metrics or is too narrowly concerned about the measures only. Boeing needs to limit such possibilities through balanced measures, independent audits, and root cause analyses. For instance, a decrease in safety measures may be viewed positively, but the actual problem might lie in the fear of reporting incidents among workers.

CHALLENGE 5: REGULATORY SKEPTICISM

The FAA and other regulators may not trust Boeing's internal reporting immediately. Boeing should treat regulatory skepticism as expected and reasonable. The company should provide transparent evidence, welcome independent review, and close corrective actions quickly.

EXPECTED PAYOFF

Payoff is not immediate recovery from financial difficulties. The payoff is a safer and more reliable production that eventually leads to financial recovery. The benefits that are to be expected from this plan are found to the right in **Figure 9**

These results are interrelated. Good training and reporting lead to better internal quality. Better internal quality will result in fewer defects and less rework. This will help to restore trust among regulators and airlines. The restored trust will facilitate reliable deliveries. Reliable deliveries will contribute to good financial performance. [Appendix C](#) describes how these results can be measured, while [Appendix D](#) lists their major implementation milestones.

CONCLUSION

Boeing's Board must support a "Safety First" Balanced Scorecard revamp. This is not simply an issue of a single aircraft type, a single vendor, a single set of missing screws, or even a single set of leaders. This is an issue of how Boeing implements its strategy

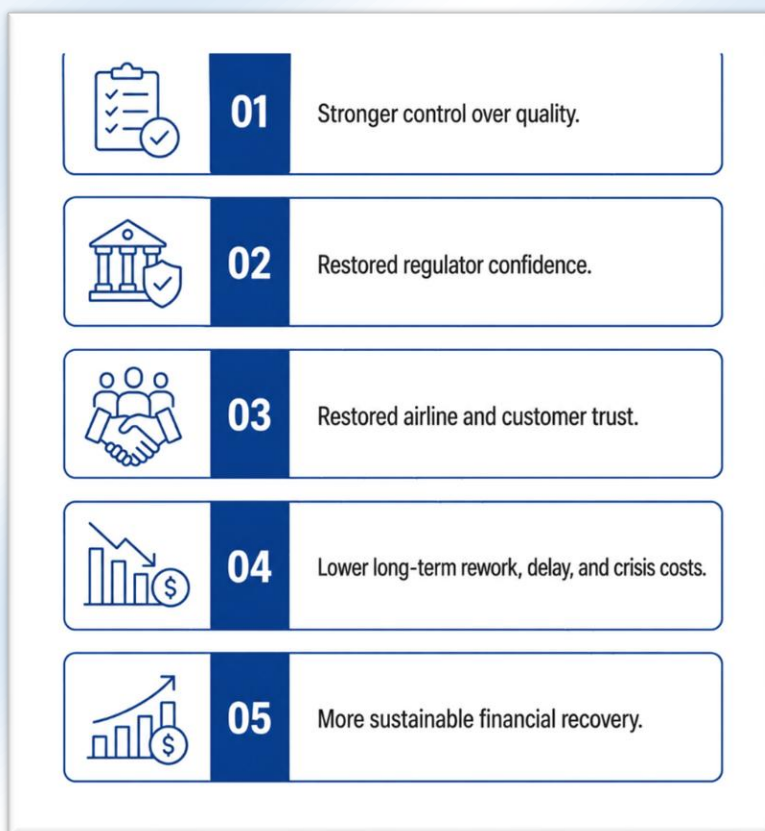


Figure 9

on a day-to-day basis. Promises will not allow Boeing to reestablish trust; instead, Boeing must demonstrate that safety and quality are a part of its leaders, vendors, employees, operations, regulators, customers, and executives.

M9: Developing a Balanced Scorecard is the proper execution approach since Boeing will be forced to link strategy to measurements, measurements to accountability, and accountability to value creation. The proper theory of value creation for Boeing is evident: safety culture creates process quality, process quality creates trust, and trust creates financial turnaround. Thus, the Board must turn the company's recovery around by using a scorecard that measures things that must change before production can be increased. The recommendation is straightforward: fix control, then scale production.

APPENDICES

APPENDIX A

This appendix supports the report’s claim that Boeing’s crisis is systemic rather than isolated.

Appendix A: Boeing Crisis Timeline and Strategic Meaning

Date/Period	Event or Indicator	Strategic Meaning
October 2018	Lion Air 737 MAX crash	Major safety failure connected to the 737 MAX crisis
March 2019	Ethiopian Airlines 737 MAX crash	Reinforced that the crisis was not a one-time event
2019–2021	Global grounding of 737 MAX	Regulatory trust and airline confidence were damaged
2021–2023	Additional reported quality issues	Suggested continuing process and quality-control weaknesses
January 2024	Alaska Airlines 737-9 door plug blowout	Showed continued manufacturing and documentation weaknesses
2024	FAA audit and increased oversight	Demonstrated that regulators did not view Boeing’s production system as fully reliable
2025	NTSB probable cause finding	Connected the Alaska incident to Boeing’s training, guidance, and oversight failures
December 2025	Boeing completed acquisition of Spirit AeroSystems	Shifted the issue from whether to reintegrate Spirit to how to execute the integration

Financial context: Boeing’s market capitalization declined significantly from 2018 to 2024, while quality and safety events damaged trust, delayed production, and increased financial pressure. Boeing’s recovery, therefore, depends on restoring safety and quality control before pursuing aggressive production growth.

APPENDIX B

Appendix B: Safety-First Strategy Map



Learning and Growth

Safety-first leadership; engineering and operations expertise; employee reporting protection; safety training completion; quality-process knowledge; stop-the-line authority



drives



Internal Processes

Stronger quality gates; Spirit integration into Boeing quality system; clear part-level accountability; standardized inspections; documentation discipline; tool-control improvement; supplier defect reduction



drives



Customer and Regulator Trust

Improved FAA audit results; stronger airline confidence; better delivery reliability; fewer defects reaching customers; clearer transparency with regulators



drives



Financial Recovery

Sustainable production increases; reduced rework and crisis costs; stronger reputation; long-term shareholder value; stable cash flow

APPENDIX C



Appendix C: Proposed Boeing Balanced Scorecard Dashboard

 PERSPECTIVE	 STRATEGIC OBJECTIVE	 KEY METRICS	 TARGET LOGIC	 OWNER
 Learning and Growth	Rebuild safety culture and employee accountability	- Safety training completion - Quality-process proficiency - Employee safety reports submitted - Reports resolved within 30 days - Non-retaliation survey score - Stop-the-line events closed	Training and reporting must improve before production growth is approved	Chief Aerospace Safety Officer; Chief Human Resources Officer; plant leadership
 Internal Processes	Improve production quality before aircraft delivery	- Supplier defect rate - Rework rate - Quality escape rate - Documentation completeness - Tool-control audit score - First-pass quality yield - Inspection pass rate	Defect and documentation trends must show sustained improvement	Chief Operations Officer; quality leadership; supplier management
 Customer and Regulator Trust	Rebuild FAA, airline, and customer confidence	- FAA audit findings - Corrective action closure rate - Airline complaints - Delivery reliability - Regulator-approved production increases - Customer confidence feedback	External trust must be proven through audit and delivery performance	CEO; regulatory affairs; commercial airplanes leadership
 Financial Recovery	Stabilize financial performance through safer production	- Rework and delay cost - Commercial deliveries after quality gates - Free cash flow - Warranty/customer compensation costs - Operating margin recovery - Backlog conversion	Financial growth should follow safety and quality proof	CFO; business unit presidents; Board committee

APPENDIX D

Appendix D: 12-Month Implementation Roadmap

	Phase 1: Control Reset	Phase 2: Quality-System Integration	Phase 3: Scorecard Governance and Production Gates	Phase 4: Institutionalize the System
PHASE				
TIMELINE	0–90 days	3–6 months	6–12 months	12–24 months
KEY ACTIONS	• Restructure safety governance • Define safety-first strategy • Create Board scorecard committee • Slow unsafe production increases • Baseline safety and quality metrics • Strengthen stop-the-line authority • Launch independent safety and quality task force	• Begin Spirit integration into one Boeing-controlled quality system • Standardize inspections and documentation • Strengthen supplier audits • Implement part-level traceability • Assign clear accountability for defect categories	• Launch full Balanced Scorecard dashboard • Report quarterly to Board • Tie production increases to quality gates • Link executive compensation to safety and quality • Communicate selected metrics to airlines and regulators	• Make scorecard part of annual planning, executive reviews, supplier management, and production decisions • Conduct independent audits • Update targets • Expand lessons across Boeing programs



Objective: Build a safe, high-quality, and resilient Boeing by embedding accountability, transparency, and continuous improvement across all operations.

APPENDIX E

Appendix E: Strategic Options Comparison

Option	Safety Impact	Quality Control	Accountability	Speed	Execution Risk	Assessment
Maintain current structure	Low	Low	Low	High	High	Fastest short-term option, but does not fix the root system
Stronger audits and contracts with Spirit separate	Moderate	Moderate	Moderate	Moderate	Moderate	Improves oversight but still leaves divided accountability
Third-party or FAA oversight as main solution	Moderate	Moderate	Moderate	Low	Moderate	Helpful for trust, but Boeing cannot outsource responsibility for its own quality system
Reintegrate Spirit within a safety-first Balanced Scorecard reset	High	High	High	Moderate	Moderate	Best long-term option because it changes the system that created repeated failures

APPENDIX F

APPENDIX F: BOARD DECISION RULE



Boeing should approve production increases only when all five conditions are met.

1	 FAA AUDIT FINDINGS AND CORRECTIVE ACTIONS FAA audit findings are reduced and corrective actions are closed on time.	 Regulatory Compliance
2	 SUPPLIER DEFECT AND REWORK RATES Supplier defect rates and rework rates show sustained improvement.	 Quality Performance
3	 SAFETY TRAINING AND QUALITY PROFICIENCY Safety training and quality-process proficiency targets are met.	 People and Capability
4	 EMPLOYEE REPORTING AND STOP-THE-LINE SYSTEMS Employee reporting and stop-the-line systems show trust and resolution.	 Culture and Accountability
5	 AIRLINE DELIVERY RELIABILITY Airline delivery reliability improves without increased quality escapes.	 Customer Reliability
 ALL FIVE CONDITIONS MUST BE MET. Production increases will be approved only when the Board has evidence of sustained improvement across all five areas.		



APPENDIX G

These models estimate the financial logic behind Boeing's safety-first Balanced Scorecard recommendation. It is not meant to be a precise forecast of Boeing's future earnings. Instead, it is a decision model showing why safety, quality, and supplier control should be treated as financial drivers. The model compares the expected impact of three scenarios: baseline recovery, safety-first recovery, and downside recovery.

The model supports a safety-first Balanced Scorecard reset because Boeing's financial recovery depends on

KEY MODEL ASSUMPTIONS				SCENARIO COMPARISON				
Assumption Category	Baseline Scenario	Safety-First Recovery Scenario	Downside Scenario	Scenario	Short-Term Financial Effect	Long-Term Financial Effect	Main Risk	Assessment
 Production approach	Gradual recovery with some continued disruption	Slower near-term ramp-up until quality gates are met	Faster production push before controls are stable	 Baseline recovery	Some improvement as production stabilizes	Financial recovery remains exposed to quality disruptions	Boeing improves but does not fully fix root execution problems	Better than doing nothing, but still risky
 Implementation cost	Moderate	High in the short term	Low in the short term					
 Defect/rework rate	Moderate	Lower over time	High	 Safety-first recovery	Higher upfront cost and slower production ramp	Lower rework, fewer delays, stronger trust, and more sustainable cash flow	Plan requires discipline and Board support	Best long-term financial option
 Delivery reliability	Moderate	Higher after quality gates stabilize	Lower					
 Regulatory confidence	Gradual improvement	Stronger improvement through evidence-based controls	Continued skepticism	 Downside recovery	Higher near-term production could improve revenue temporarily	More defects, more rework, more regulatory limits, and greater crisis risk	Boeing repeats the production-before-control problem	Highest long-term risk
 Customer trust	Slow recovery	Stronger recovery over time	Weak recovery					
 Financial risk	Moderate	Lower long-term risk	High					

FINANCIAL IMPACT LOGIC			SENSITIVITY ANALYSIS			
Financial Driver	How It Affects Boeing	Balanced Scorecard Connection	Variable Tested	Low Impact Case	High Impact Case	Meaning for Recommendation
 Rework cost	More defects increase labor, parts, inspection, and delay costs	Internal Processes	 Defect/rework rate	Small reduction in defects	Major reduction in defects	The more defects fall, the stronger the financial case for the safety-first plan
 Delivery delays	Delayed aircraft reduce cash flow and customer satisfaction	Customer/Regulator Trust and Financial	 Average rework cost per aircraft	Rework costs are limited	Rework costs are large	Higher rework costs make prevention more valuable
 Regulatory restrictions	FAA limits can slow production and deliveries	Customer/Regulator Trust	 Length of production delay	Short delays	Long delays	Longer delays make quality gates more important because uncontrolled production can trigger larger disruption
 Supplier defects	Poor supplier quality increases final assembly disruption	Internal Processes	 FAA approval timing	Faster approval	Slower approval	Boeing benefits financially if regulators trust its evidence and allow disciplined production increases
 Employee reporting	Strong reporting catches issues earlier and reduces crisis costs	Learning and Growth	 Implementation cost	Lower cost to implement safety reset	Higher cost to implement safety reset	Even with higher upfront cost, the plan is justified if it prevents larger future disruption
 Customer trust	Stronger airline confidence supports future orders and delivery acceptance	Customer/Regulator Trust	 Customer trust recovery	Airlines remain cautious	Airlines regain confidence faster	Faster trust recovery improves delivery reliability and future order strength
 Crisis costs	Major safety failures create legal, reputational, and operational costs	Financial				



Safety. Quality. Accountability. Performance.

operational control. A production-first approach may look better in the short term, but it creates greater risk if quality escapes, rework, delays, and regulatory restrictions continue. The safety-first option is financially stronger because it reduces the root causes of disruption. The key conclusion is that safety and quality are not only ethical or regulatory requirements for Boeing; they are financial recovery drivers.

Production growth should be treated as a reward for system readiness, not as a substitute for it.

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